CS230: Requirements Engineering
School of Information Technology, Engineering, Mathematics and Physics

Final Examination
Semester 1 2022

F2F Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

Total marks: 100

Instructions:

1. *This exam three sections, with a total of 10 questions.*
2. *Section A 30 marks, Section B 40 marks, Section C 30 marks.*
3. *This exam is worth 50 % of your overall mark.*
4. *The minimum pass mark for this exam is 40 out of 100 marks*
5. *There are 15 pages in this paper including this cover page.*
6. *This is a closed book Exam*
7. *Start each question on a new page in your answer booklet.*
8. *All the questions are compulsory. There are no choices.*

Student Name: _____

Student ID: _____

~ALL THE BEST~

Section A: Short Answers (30 Marks)

Question 1 **(5 marks)**

a) What is Requirements Engineering? (1 mark)

The process of establishing the services that the customer requires from a system and the constraints under which it operates and is developed.

Mark Allocation: correctly answered = full mark. Partial correct will get 50%.

b) How is a functional requirement different from a non-functional requirement? (1 marks)

Functional requirements define what the system should do.

Non-functional requirements are not directly concerned with specific system functions but specify required system properties or place constraints on the system or its development process.

OR

When a functional requirement is not validated the system will still be operational but when a NFR is not validated the system will not be operational and could fail leading to catastrophic consequences.

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

c) Domain requirements are constraints on the system from the domain of operation. What are the two domain requirements problems? (2 marks)

Understandability - Requirements are expressed in the language of the application domain. This is often not understood by software engineers developing the system.

Implicitness - Domain specialists understand the area so well that they do not think of making the domain requirements explicit. This is major problem in communications.

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

d) List any two elements of Requirements Analysis? (1 mark)

- Scenario based models – use case and user stories
- Class model – class diagram
- Behavioral models – state diagram
- Flow model – DFDs

- Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

Question 2 **(5 marks)**

a) Scenarios are descriptions of how a system is used in practice. They are helpful in requirements elicitation as people can relate to these more readily than abstract statement of what they require from a system. Give any two items which should be included in a scenario. (2 marks)

- A description of what's expected when the scenario starts.
- A description of the normal flow of events.
- A description of what can go wrong and how to handle it.
- Information about concurrent activities.
- A description of the system state when the scenario finishes.

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

b) What is the main benefit of formal methods? (1 mark)

Their principal benefits are in reducing the number of errors in systems so their main area of applicability is critical systems

Mark Allocation: correctly answered = full mark. Partial correct will get 50%.

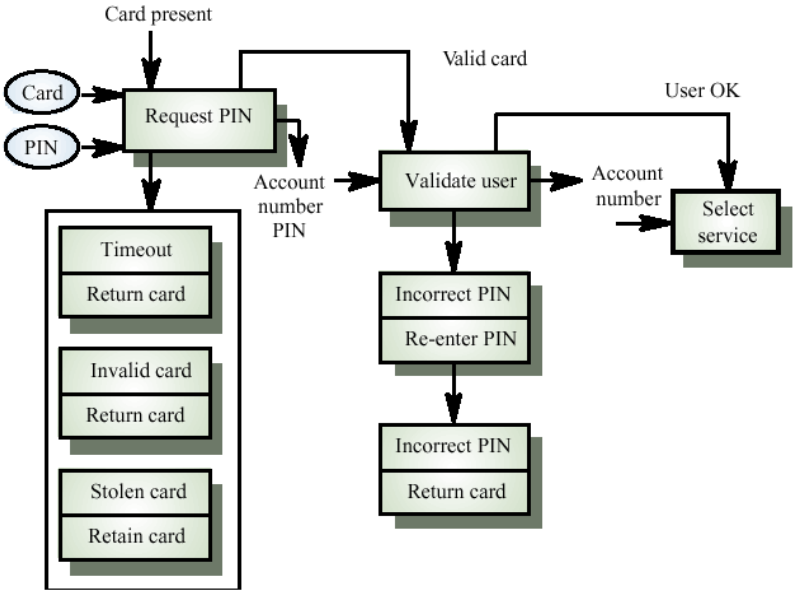
e) Explain the term Algebraic and Model-based approach in Specification techniques? (2 marks)

- Algebraic approach
  - The system is specified in terms of its operations and their relationships
- Model-based approach
  - The system is specified in terms of a state model that is constructed using mathematical constructs such as sets and sequences. Operations are defined by modifications to the system's state

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

Question 3 **(6 marks)**

a) Jone wants to change his ATM card's pin. Consider the event scenario below to answer the questions.



List any two exceptions descriptions in this event. (2 marks)

- Timeout. Customer fails to enter a PIN within the allowed time limit
- Invalid card. The card is not recognised and is returned
- Stolen card. The card has been registered as stolen and is retained by the machine

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

Differentiate between Requirements Verification and Validation? (2 marks)

- Requirements Validation
  - **Check that the right product is being built**
  - Ensures that the software being developed (or changed) will satisfy its stakeholders
  - Checks the software requirements specification against stakeholders goals and requirements
- Requirements Verification
  - **Check that product is being built right**
  - Ensures that each step followed in the process of building the software yields the right products
  - Checks consistency of the software requirements specification artefacts and other software development products (design, implementation, ...) against the specification

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

b) Peter's company has written several requirements for the new Inventory system for a supermarket. He is tasked to verify and validate the requirements. List and explain any two Requirements Verification and Validation techniques he can use. (2 marks)

- Simple checks
  - Traceability, well-written requirements

- Prototyping
- Functional test design
- User manual development
- Reviews and inspections
    - Walkthroughs
    - Formal inspections
    - Checklists
- Model-Based V&V
    - First-order logic
    - Behavioral models

Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

Question 4 **(9 marks)**

Mark Allocation: both correctly answered = full mark. No partial mark for Q4

a) There are many information acquisition techniques using to gather the information about current system. It is very important to plan information acquisition well for different objective. For the following objectives, state the best technique to apply. (2 marks)

Objective	Technique
To know about the background of the company, their aim, objectives and current system.	Background reading
To understand the roles of each departments, line managers and team structure in each department.	Interview
To follow up development of business understanding and see physical processes.	Observation
To collect information from a large number employees on a common topic.	Questionnaire

b) What does Abstraction and Modeling, Decomposition and Hierarchy solve? (1 mark)
complexity / Complexity

c) Write the type of hierarchy is this "University organizational structure: the VC is superior to the Head of Schools, who are superior to their Discipline Coordinators who are superior to their teaching staff". (1 mark)
Part-of

d) Difference between Inception and Elaboration? (1 mark)

Elaboration focuses requirements capture and determining the structure of the system. Inception is concerned with determining the scope and purpose of the project.

e) What is the difference between requirement and specification? (1 marks)

- Requirement – understanding between customer and supplier
- Specification – what the software must do

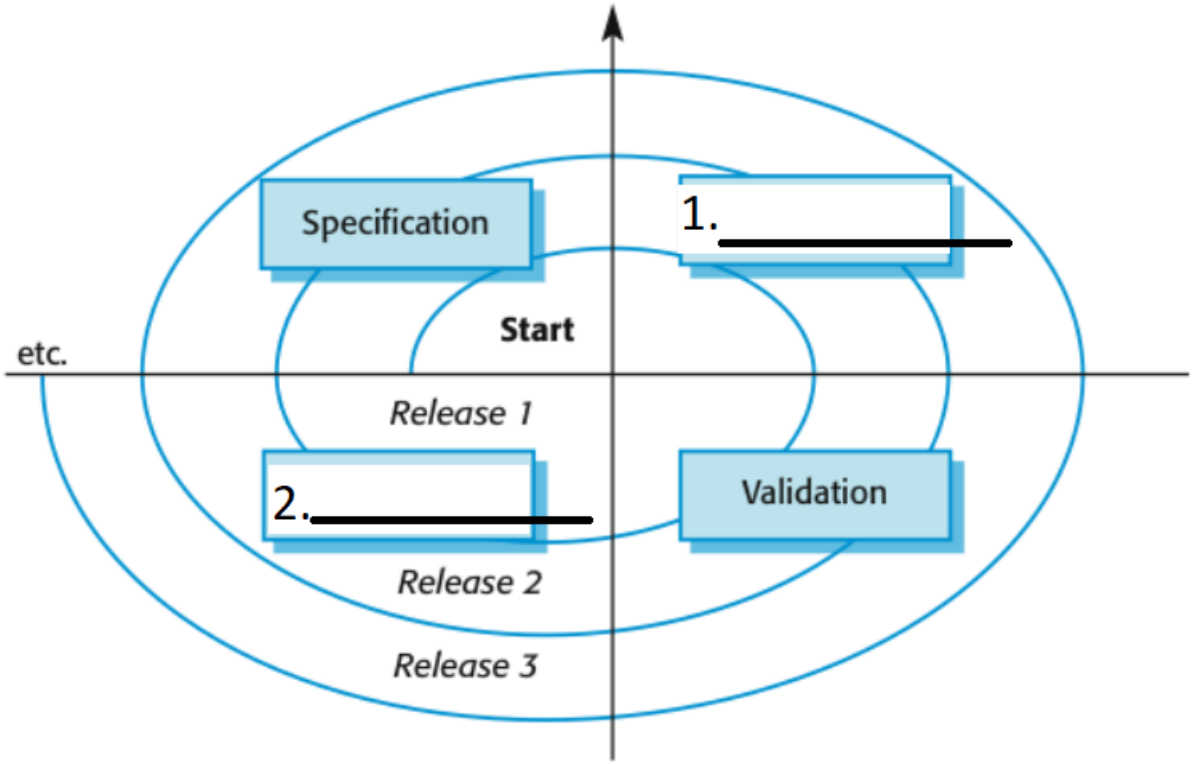
f) List and explain three Characteristics of a Good SRS. (3 marks)

1. Unambiguous
2. Complete
3. Verifiable
4. Consistent
5. Modifiable
6. Traceable
7. Usable during the Operation and Maintenance Phase

Question 5 **(5 marks)**

Mark Allocation: both correctly answered = full mark. No partial mark for Q5

a) Write the missing process from the spiral model of evolution at space provided at 1 and 2. (2 marks)



At 1: implementation

At 2: operation

b) Bula IT developed new software Bula Hangout and does continuous changes over time. Give any three reasons why software changes are inevitable. (3 marks)

- New requirements emerge when the software is used;
- The business environment changes;
- Errors must be repaired;
- New computers and equipment is added to the system;
- The performance or reliability of the system may have to be improved.

Section B: Short Answers (30 Marks)

Refer to the following case study to answer questions 6.

Online Hypermarket Dispatch System

The purpose of the Online Hypermarket Dispatch System is to provide ordering and delivery logistic for the hypermarket.

Initially, prospective customers place an order request for selected items. After order is received, the salesperson checks the inventory for the availability of items on the order. To verify this, the salesperson has to contact the dispatch clerk in the warehouse. If there is not enough stock, the salesperson cancels the order. However, if there are enough stock to meet the order, the salesperson then requests the credit card information and at the same time the shipping address from the customer. The shipping details are then passed on to the dispatch clerk at the warehouse who then packs the order and ships it to the address.

For delivery purposes, each shipment is assigned to a delivery person. The delivery person is responsible to deliver the package to the given address.

Question 6 **(10 marks)**

a) Identify and List the use cases with their brief descriptions. (3 marks)

Actors:

Buyer/Customer, Sales, Warehouse

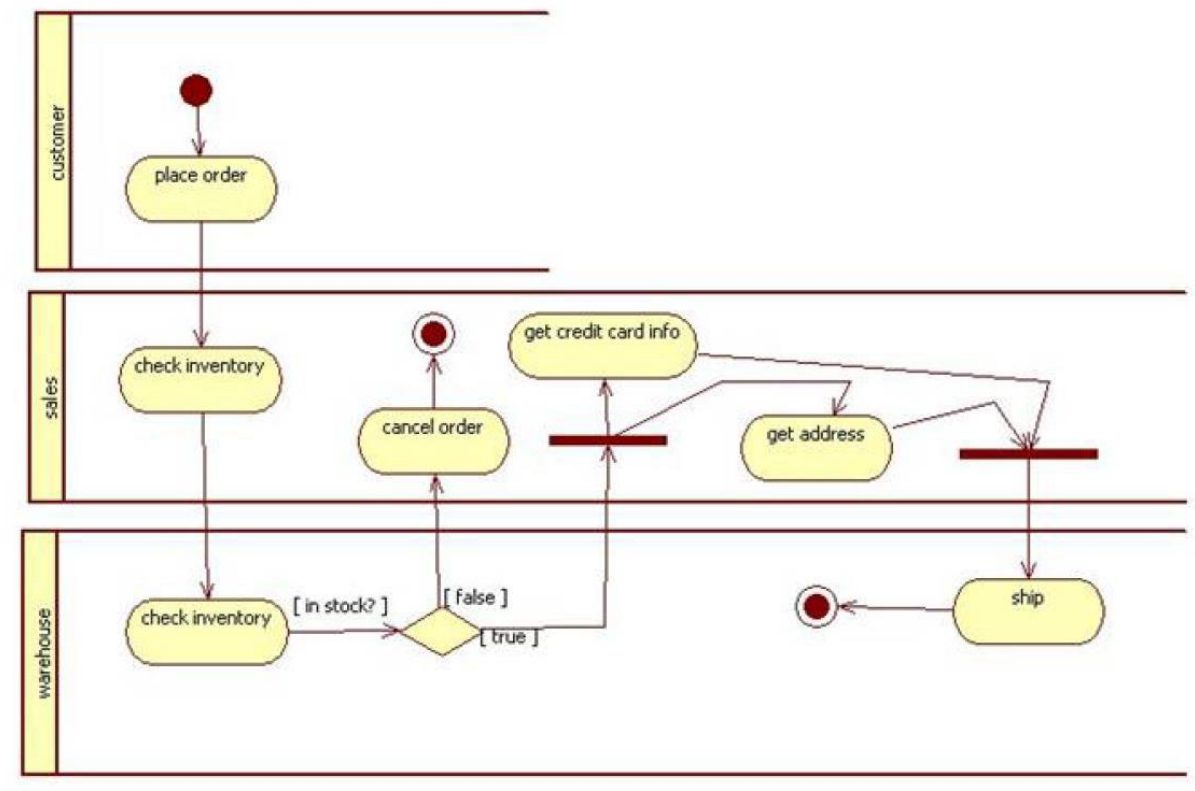
Use Cases

User	User goal & Resulting Use Case
Buyer	Place order
Sales	Check inventory Cancel order Request credit card info and address
Warehouse	Check inventory Inform sales about stock Ship the order

Mark Allocation:

- Identified all users (1.5 marks)
- All use case identified (1.5 marks)

b) Develop an Activity diagram based on the above case study. Note any ambiguities or questions that you have as you develop the model. If you need to make assumptions, also note them. (7 marks)



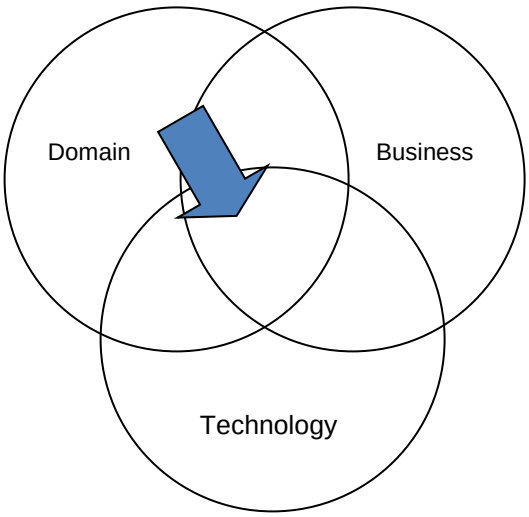
Mark Allocation:

- Identified all activities correctly (3 marks)
- All the data flow correctly (2 marks)
- Correct decision flow (1 mark)
- Correct activity diagram structure and boxes (1 mark)

Question 7 Problem Domain and Modelling (5 Marks)

With a diagram illustrate and explain the key factors of the Domain-Specific Software Engineering. (3 marks)

- Domain
- Technology
- Business



Domain + Business + Technology

"Domain-specific software engineering"

Applies technology to domain-specific goals, tempered by business and market knowledge

Mark Allocation: three correctly answered + digram = full mark. Partial ones correct (1/2 each)

Explain the term Algebraic and Model-based approach in Specification techniques? (2 marks)

- Algebraic approach
  - The system is specified in terms of its operations and their relationships
- Model-based approach
  - The system is specified in terms of a state model that is constructed using mathematical constructs such as sets and sequences. Operations are defined by modifications to the system's state
- Mark Allocation: both correctly answered = full mark. Partial or one correct will get 50%.

Question 7 Software evolution (7 Marks)

Program evolution dynamics is the study of the processes of system change. After major empirical studies, Lehman and Belady proposed that there were a number of 'laws' which applied to all systems as they evolved.

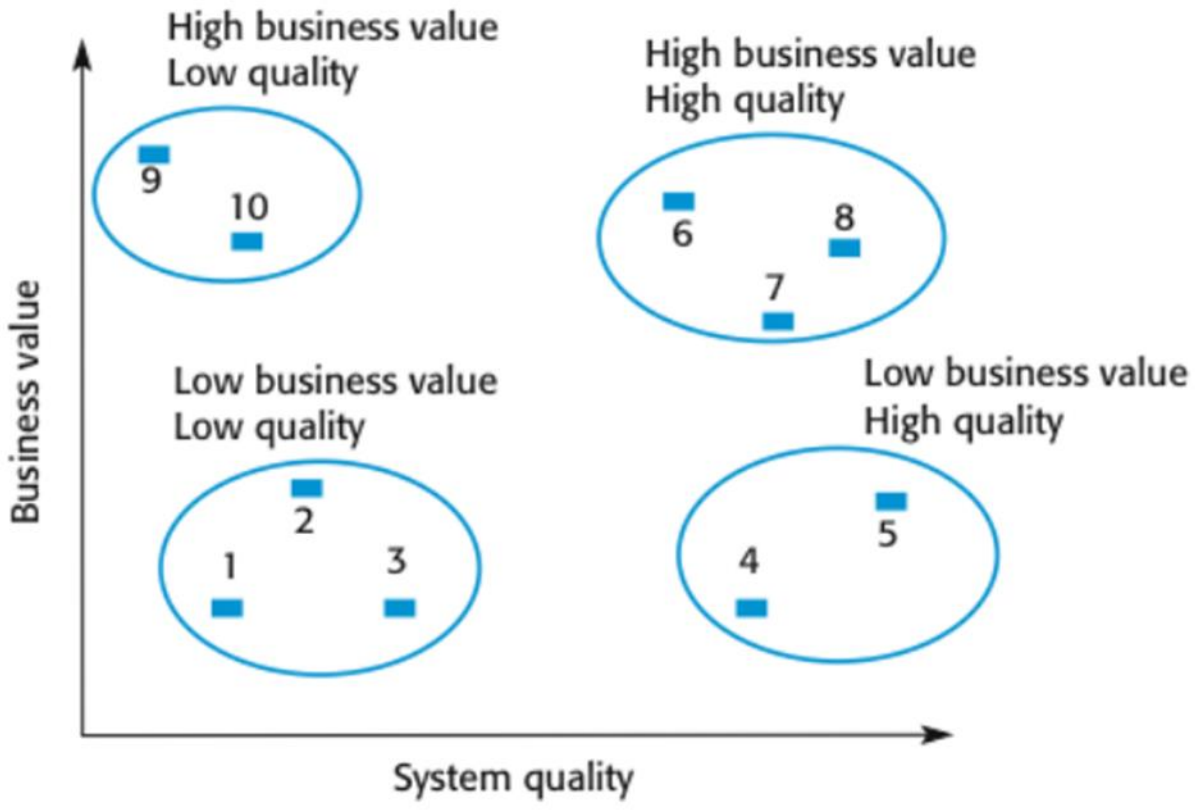
List and describe any three Lehman's law. (3 marks)

Law	Description
Continuing change	A program that is used in a real-world environment must necessarily change, or else become progressively less useful in that environment.
Increasing complexity	As an evolving program changes, its structure tends to become more complex. Extra resources must be devoted to preserving and simplifying the structure.
Large program evolution	Program evolution is a self-regulating process. System attributes such as size, time between releases, and the number of reported errors is approximately invariant for each system release.
Organizational stability	Over a program's lifetime, its rate of development is approximately constant and independent of the resources devoted to system development.

Law	Description
Conservation of familiarity	Over the lifetime of a system, the incremental change in each release is approximately constant.
Continuing growth	The functionality offered by systems has to continually increase to maintain user satisfaction.
Declining quality	The quality of systems will decline unless they are modified to reflect changes in their operational environment.
Feedback system	Evolution processes incorporate multiagent, multiloop feedback systems and you have to treat them as feedback systems to achieve significant product improvement.

Mark Allocation: three correctly answered = full mark. Partial or correct ones get 1 mark each

1. Organizations that rely on legacy systems must choose a strategy for evolving these systems. The chosen strategy should depend on the system quality and its business value. Refer to system quality and its business value given below and explain the four outputs while dealing with an organization? (4 marks)



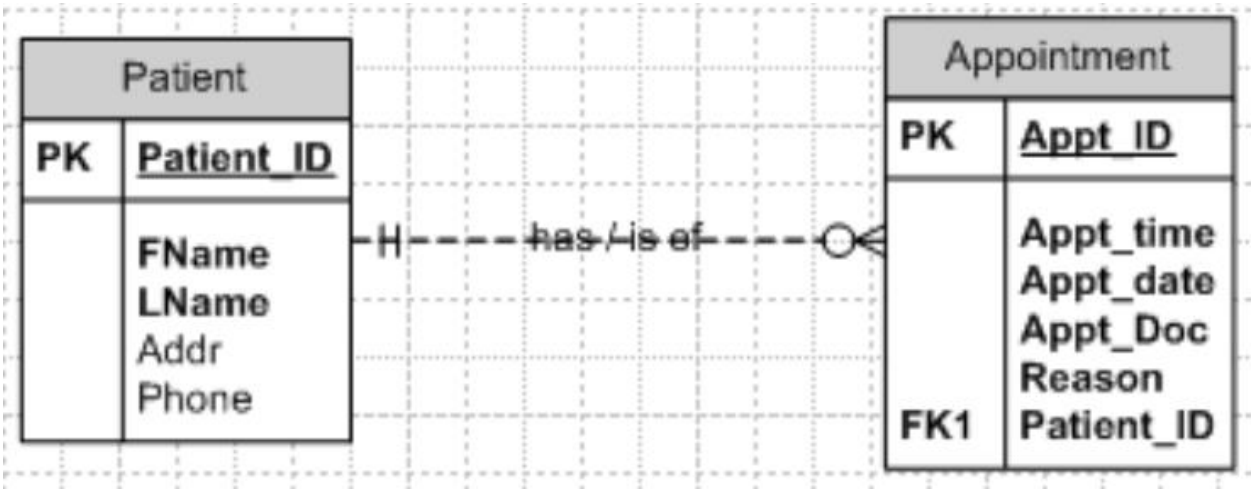
Solution:

- Low quality, low business value: **should be scrapped.**
- Low-quality, high-business value: **make an important business contribution but are expensive to maintain. Should be re-engineered or replaced if a suitable system is available.**
- High-quality, low-business value: **replace with COTS, scrap completely, or maintain.**
- High-quality, high business value: **continue in operation using normal system maintenance.**

Mark Allocation: four correctly answered = full mark. Partial or correct ones get 1 mark each

Question 8 ERD (7 marks)

Find below a diagram representing the classes with their associations for a Medical healthcare appointment system. Explain the associations between both patient and appointment (2 marks)



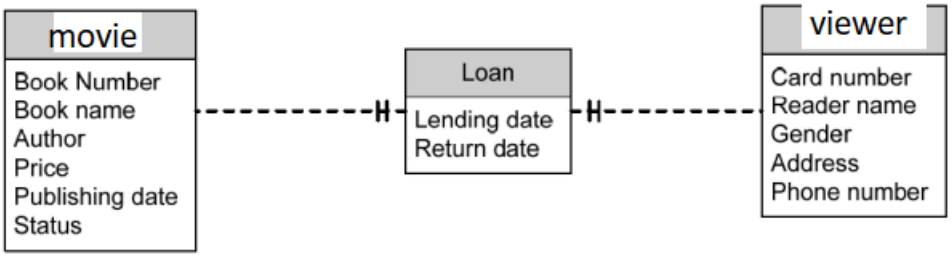
- Any one patient (Bill Jones, for example) may have no appointments but may have many.
- Any one appointment (#2191 at 4 p.m. on 5/21) must be for one patient and no more than one

Mark Allocation: both correctly answered = full mark. Partial or correct ones get 1 mark each

A Bulaflix has many DVD movies and many subscribers/viewers. One viewer can borrow many movies. One movie can be borrowed by many readers at different time. If a movie is lent to a viewer, the lending date and return date must also be recorded. Attributes about movie and viewer are given below:

- Movie: movie number, movie name, director, price, released date, status;
- Viewer: card number, viewer name, gender, address, phone number.

Draw an entity relationship diagram for the above case. (5 marks)



Mark Allocation:

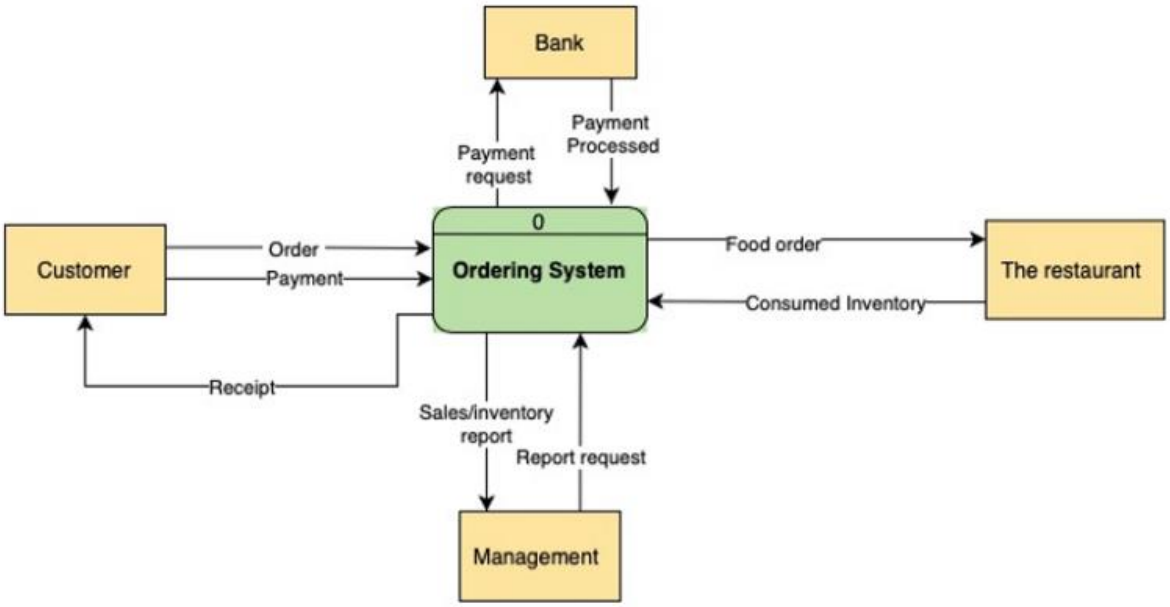
- Identified all class correctly (3 marks)
- All the relationship and cardinality correctly (2 marks)
- Partial marks for the ones correct

Question 9 UML

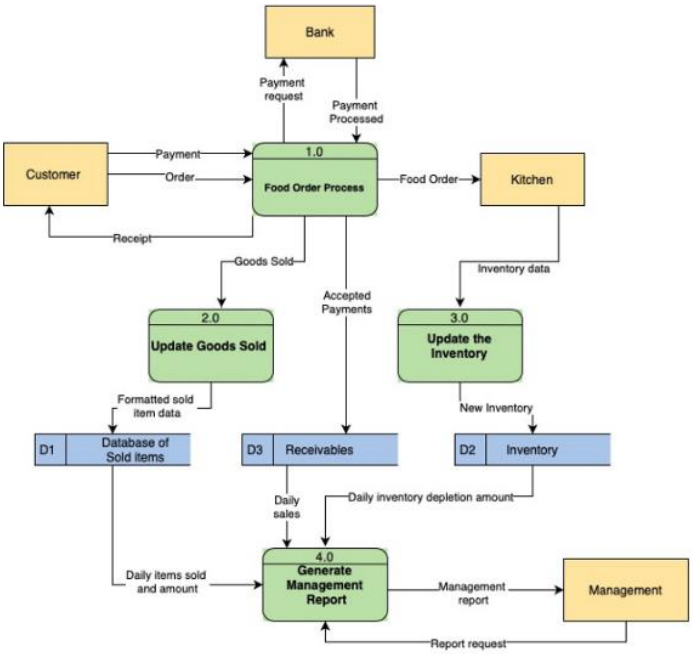
The Context Diagram represents the entire information system as one big process.

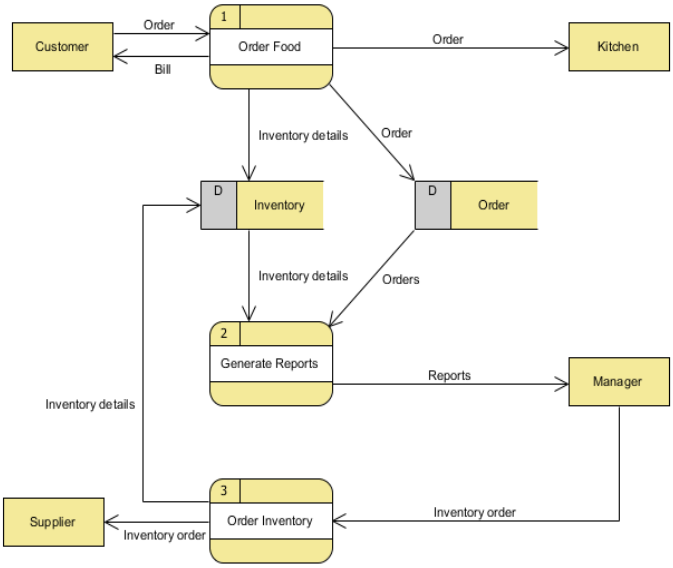
However, any information system will have many processes. The Level 0 DFD shows the major processes of the information system.

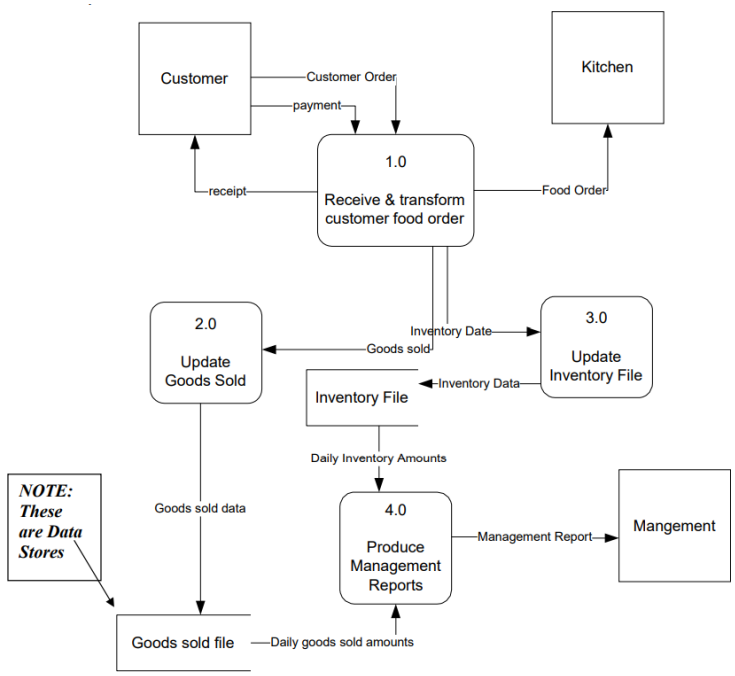
A simple context diagram for a 'Pizza ordering system' is given below. Draw a DFD level 0 diagram for the system. **(7 marks)**



Answer


Bank
Payment request
Payment Processed
Customer
Payment
Order
1.0
Food Order Process
Food Order
Kitchen
Receipt
Goods Sold
Inventory data
Accepted Payments
2.0
Update Goods Sold
3.0
Update the Inventory
Formatted sold item data
New Inventory
D1
Database of Sold items
D3
Receivables
D2
Inventory
Daily sales
Daily inventory depletion amount
4.0
Generate Management Report
Daily items sold and amount
Management report
Management
Report request


Customer
Order
Bill
1
Order Food
Order
Kitchen
Inventory details
Order
D
Inventory
D
Order
Inventory details
Orders
2
Generate Reports
Reports
Manager
Inventory details
3
Order Inventory
Supplier
Inventory order
Inventory order



Mark Allocation:

- Identified all objects correctly (3 marks)
- All dataflow correct (4 marks)
- Correct labelling (1 mark)
- Partial marks for the ones correct

Question 10 Requirements Elicitation (4 marks)

FFW Fiji Fashion Week System

Fiji Fashion Week shows have become an important information source for new products, new fashions, and new fabrics. In addition to the large providers of outdoor clothing and fabrics, there are many smaller providers (Women in SMEs). It is important for FFW to capture information about these suppliers while the Fashion Week is in progress. It is also important to obtain information about specific merchandise products that FFW plans to purchase.

Additionally, if quality photographs of the products can be obtained while at the Fashion Week, then the creation of online product pages is greatly facilitated. It is recommended that a new system be developed and deployed so field purchasing agents can communicate more rapidly with the retail shops and markets about suppliers and specific products/fabrics of interest. This system should be deployed on portable equipment.

The new system should be capable of:

- Collecting and storing information about the manufacturer/wholesaler (suppliers)
- Collecting and storing information about sales representatives and other key personnel for each supplier

- Collecting information about products
- Taking pictures of products (and/or uploading stock images of products)
- Functioning as a stand-alone without connection
- Connecting via Wi-Fi (Internet) and transmitting data
- Connecting via telephone and transmitting data

A) Which two elicitation technique(s) will you us to elicit the requirements of this system? Why? (2 marks)

- Open interviews that target different types of families in the area. People in the area are sensitive with private information so you will need to go around it and not stick a firm set of questions.
- Open/Close interviews with financial experts.
- Online Questionnaires: They help with forming a rough idea of how people manage their budget. The software is for general use so you need to know the trend.
- Workshops: where you get financial experts, and normal people, as well as project managers together. If project budget allows, this is a perfect technique.
- Prototyping: Such a system is highly interactive, so there will be need to focus on its interface aspects.

Mark Allocation: both correctly answered = full mark. Partial or correct ones get 1 mark each

B) Identify at least 2 stakeholders involved. Justify your choices. (2 marks)

- Domain experts: Fashion week designers
- Manufacturers including Women and SME – they will provide details about the products
- Field purchasing agents
- Retail shops and markets– they will see the products of their interest
- Authority including the sponsors – Ministry of social affairs, etc.
- Development team

Mark Allocation: both correctly answered = full mark. Partial or correct ones get 1 mark each

Section C: Case Study (30 Marks)

Question 9 Scenario based questions.

Read the problem statement given below very carefully and use it to answer the questions that follow.

You are required to read the following system specifications and answer the questions below.

BulaClothingfj mobile app

The BulaClothingfj mobile app is a type of e-business exchange that does business entirely on mobile app via the Internet. The company sets a platform for buyers and sellers of used clothes.

To offer clothes for sale, a person must register with BulaClothingfj. The person must provide a current physical address, postal address, contact details as well as a current valid e-mail address. The system sends verification email to have email and once verified then the system open account for this person. The system is accessed by the seller only through a secured and authenticated portal.

A seller can list clothes on the system through a special form on the app. The form asks for all the pertinent information about the clothes: its category, type, size, its general condition, and the asking price. A seller may list as many clothes as desired. The system maintains an index of all clothes in the system so buyers can use the search engine to search for clothes. The search engine allows searches by type, seller, category, and keyword.

People who want to buy the clothes goes to the app and search for the clothes they looking for. If they decide to buy anything they have to register an account with a credit card to pay for the clothes. The mobile app communicates all this information on secure servers.

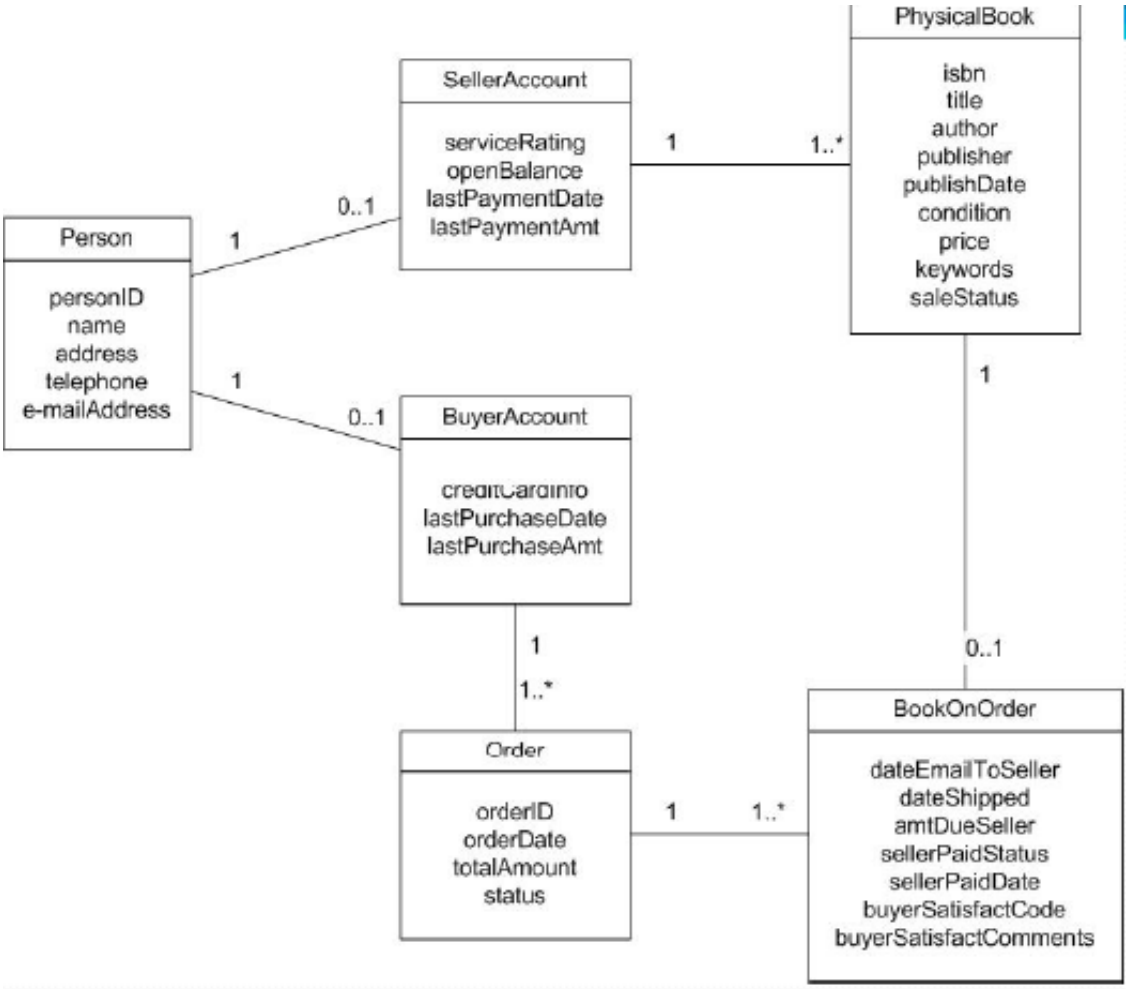
When a purchase is made, BulaClothingfj mobile app sends an e-mail notification to the seller of the clothes about the purchased clothes as well as payment information. It also marks that clothes/items as sold. The app maintains and displays an open order until it receives notice that the clothes has been shipped. After the seller receives notice that a listed clothes has been sold, the seller must notify the buyer via e-mail within 48 hours that the purchase is noted. Shipment of the order must be made within 24 hours after the seller sends the notification e-mail. The seller sends a notification to the buyer and BulaClothingfj when the shipment is made.

After receiving the notice of shipment, BulaClothingfj maintains and displays the order in a shipped status. At the end of each month, a check is mailed to each seller for the clothes orders that have remained in a shipped status for 30 days. The 30-day waiting period exists to allow the buyer to notify BulaClothingfj if the shipment doesn't arrive for some reason or if the clothes isn't in the same condition as advertised.

If they want, buyers can enter a service code for the seller. The service code is an indication of how well the seller is servicing clothes purchases. Some sellers are very active and use BulaClothingfj as a major outlet for selling clothes. Thus, a service code is an important indicator to potential buyers.

For same BulaClothingfj mobile app case, develop the diagrams in (a) to (d):

a) Convert your ERD to a domain class diagram (7 marks)

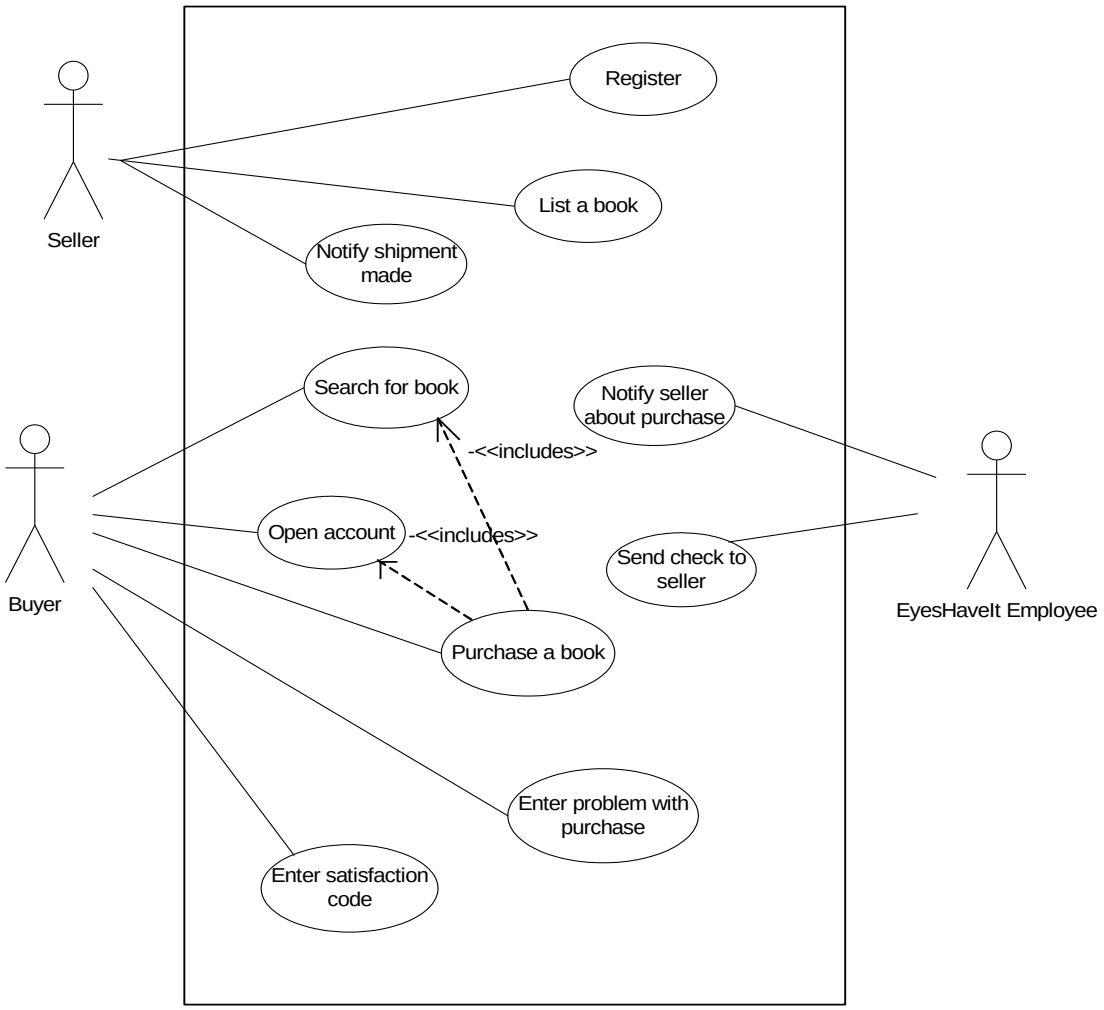


Mark Allocation:

- Identified all classes correctly (3 marks)
- All relationship and cardinality (3 marks)
- Correct labelling and (1 mark)
- Partial marks for the ones correct

---

b) Create a detailed use case diagram for the BulaClothingfj mobile app capturing the functional requirements. (10 marks)



Mark Allocation:

- Identified all use cases correctly (3 marks)
- All relationship and arrows (3 marks)
- Identified all actors (3 marks)
- Correct system rectangle and actors outside (1 marks)
- Partial marks for the ones correct

---

c) Using the above description, write a fully developed description for use cases: **Add a seller**, use case scenario showing the Brief Description, Actors and the Flow of Events (FoE)/Activities, preconditions, postconditions exception conditions etc. For the FoE show the Actor and System tasks. (7 marks)

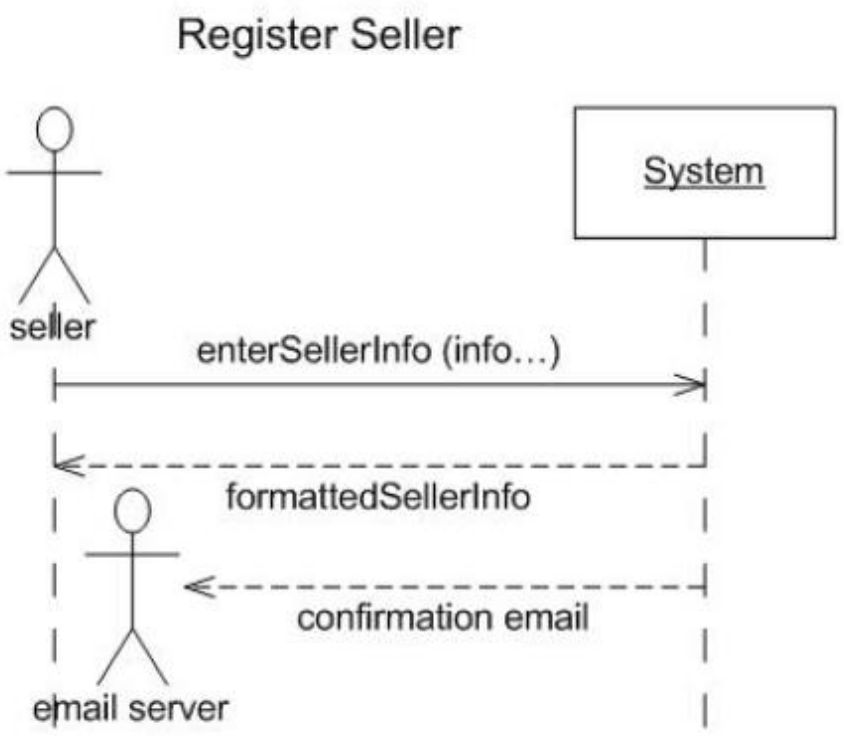
Use Case Name:	Register	
Scenario:	Register/add a new seller	
Triggering Event:	A new seller wants to sell books.	
Brief Description:	Seller decides he/she would like to list a book(s). Seller registers and receives a confirmation e-mail.	
Actors:	Seller E-mail server	
Stakeholders:		
Preconditions:	Seller must not exist in the system. Seller must have all information necessary to register.	
Postconditions:	Seller has an account to list books.	
Flow of Activities:	Actor	System
	1. Seller connects to EyesHaveIt.com and fills out registration form. 2. Seller submits registration form.	2.1 System notifies seller that a confirmation e-mail will be sent. 2.2 System e-mails confirmation of registration to seller via the Email Server.
Exception Conditions:	1.1 If the seller already exists in the system, the system sends prior login and password to e-mail address. 2.1 If the seller was removed from the system for bad transactions/credit, the system sends the seller an e-mail notifying the seller of the situation and no account is created.	

Mark Allocation:

- Correct flow (2)

- Precondition, postcondition and exception condition (3)
- Trigger (1)
- Actor and description (1)
- Partial marks for correct bits

d) Now create a System Sequence diagram for the scenario you have given in part (c) above: **Add a seller**. (6 marks)



Mark Allocation:

- Correct sequences and labels (3)
- Correct actor and system (2)
- Correct arrows (1)
- Partial marks for the ones correct

THE END